CS413
Problem Solving & Decision Making

Section 1

Problem Solving & Decision Making

- Course overview
- Grading
- Intro to Queuing Theory

Problem Solving & Decision Making

- Course overview:
  - Identifying and defining problems
  - Analyzing Problems
  - Working with Problem Owners and Stakeholders
  - Developing Effective Problem Statements
  - Simplifying Complex Problems
  - Identifying and Managing Risks
  - Problem Solving Strategies
  - **Queuing Theory**
  - **Inventory Theory**
- Labs will discuss queuing theory and inventory theory in practical detail.

Problem Solving & Decision Making

- Grading:
  - Three Quizzes – counting the highest two grades
  - Tasks / Project
  - Practical Exam

Introduction (1/6)

Queuing Theory:

- Queues (or waiting lines) help facilities or businesses provide service in an orderly fashion.
- In queuing theory, we discuss mathematical models that represent the process by which queues operate.
- The ultimate objective of the analysis of queueing systems is to understand the behavior of their underlying processes so that informed and intelligent decisions can be made in their management.

Introduction (2/6)

Terminology:

- We identify the unit demanding service, whether it is human or otherwise, as ***customer***.
- The unit providing service is known as the ***server***.
- This terminology of customers and servers is used in a generic sense regardless of the nature of the physical context.

Introduction (3/6)

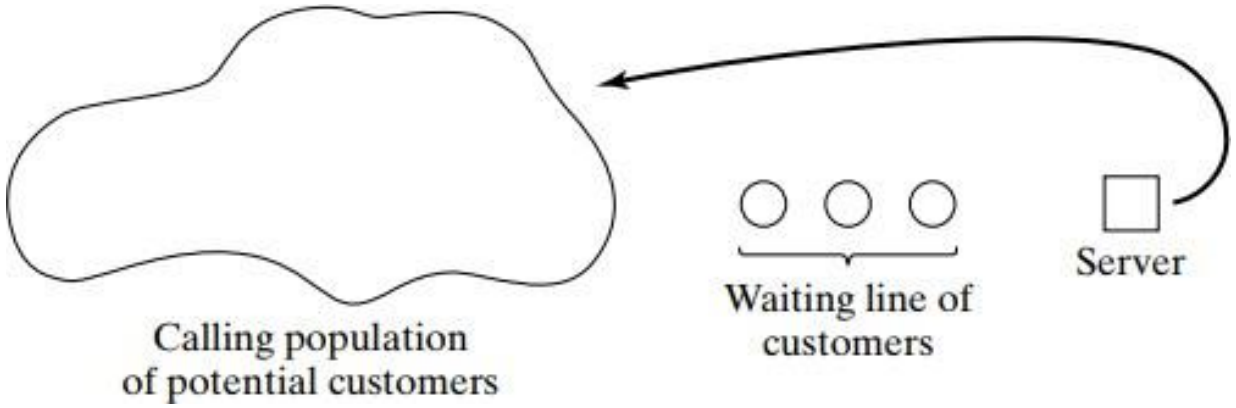
Examples:

- Patients arrive at a doctor's clinic for treatment.
- Clients waiting in line to speak with a bank teller.
- In communication systems, voice or data traffic queue up for lines for transmission.
- When accessing a webpage, a request is sent from the computer to the website's hosting computer.

Introduction (4/6)

Queueing Model:

- Simulation is often used in the analysis of queueing models. In a simple but typical queueing model, customers arrive from time to time and join a queue (waiting line), are eventually served, and finally leave the system.



Introduction (5/6)

- Queueing models, whether solved mathematically or analyzed through simulation, provide the analyst with a powerful tool for designing and evaluating the performance of queueing systems.
- Typical measures of system performance include server utilization (percentage of time a server is busy), length of waiting lines, and delays of customers.

Introduction (6/6)

- Quite often, when designing to improve a queueing system, the analyst (or decision maker) is involved in tradeoffs between server utilization and customer satisfaction in terms of line lengths and delays.

Description of Queuing Sys (1/8)

- The key elements of a queueing system are the ***customers*** and ***servers***.
- The term customer can refer to people, machines, trucks, mechanics, patients, airplanes, e-mail, cases, or orders—anything that arrives at a facility and requires service.
- The term server might refer to receptionists, repair personnel, mechanics, medical personnel, runways at an airport, automatic packers, order pickers, or CPUs in a computer—any resource (person, machine, etc.) that provides the requested service.

Description of Queuing Sys (2/8)

Queueing System:

1. The Calling Population.
2. System Capacity.
3. The Arrival Process.
4. Queue Behavior and Queue Discipline.
5. Service Times and the Service Mechanism.

Description of Queuing Sys (3/8)

1. The Calling Population (1/3):

- Also called: Input source (source of customers).
- The population of potential customers, referred to as the *calling population*, may be assumed to be finite or infinite.

Description of Queuing Sys (3/8)

1. The Calling Population (2/3):

- Infinite size:
  - Simulation assumptions: If a unit leaves the calling population and joins the waiting line or enters service, there is ***no change*** in the arrival rate of other units that may need service.
  - Most simulated models.
- Finite size:
  - Less common and complicates the simulation.

Description of Queuing Sys (3/8)

1. The Calling Population (3/3):

- The main difference between finite and infinite population models is how the arrival rate is defined. In an infinite population model, the arrival rate (i.e., the average number of arrivals per unit of time) is not affected by the number of customers who have left the calling population and joined the queueing system.

Description of Queuing Sys (4/8)

2. System Capacity:

- Is the maximum number of units that can be accommodated in the system.
- In many queueing systems, there is a limit to the number of customers that may be in the waiting line or system.
- It's unlimited capacity if the system can accommodate any number of units in the waiting line.

Description of Queuing Sys (5/8)

3. The Arrival Process:

- The arrival process for infinite-population models is usually characterized in terms of interarrival times of successive customers. Arrivals may occur at scheduled times or at random times. When at random times, the interarrival times are usually characterized by a probability distribution.

Description of Queuing Sys (6/8)

4. Queue Behavior and Queue Discipline (1/2):

- *Queue behavior* refers to the actions of customers while in a queue waiting for service to begin. In some situations, there is a possibility that incoming customers will balk (leave when they see that the line is too long), renege (leave after being in the line when they see that the line is moving too slowly), or jockey (move from one line to another if they think they have chosen a slow line).

Description of Queuing Sys (6/8)

4. Queue Behavior and Queue Discipline (2/2):

- *Queue discipline* refers to the logical ordering of customers in a queue and determines which customer will be chosen for service when a server becomes free.
- Common queue disciplines include first-in-first-out (FIFO), last-in-first-out (LIFO), service in random order (SIRO), shortest processing time first (SPT), and service according to priority (PR).

Description of Queuing Sys (7/8)

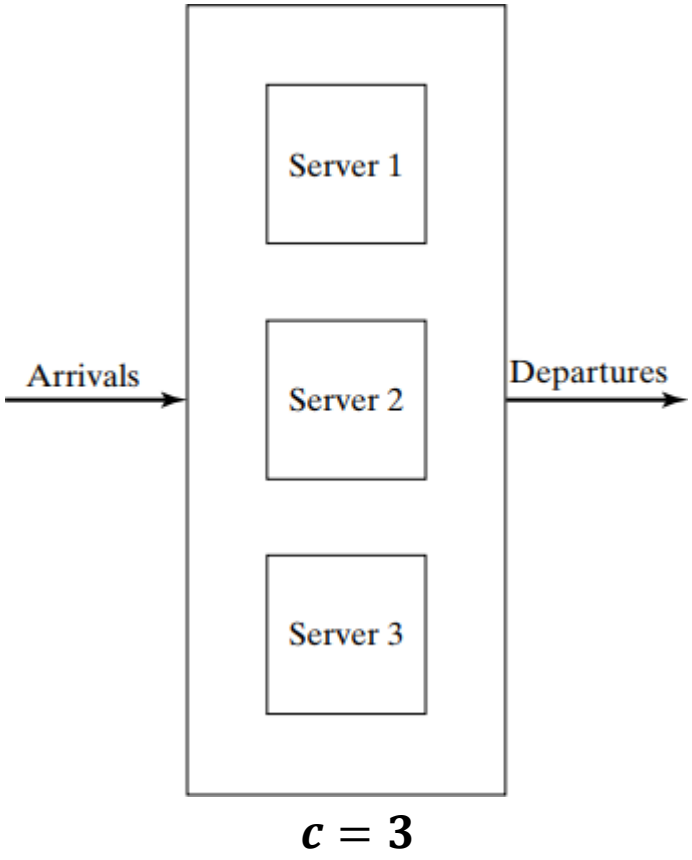
5. Service Times and the Service Mechanism (1/3):

- The service times of successive arrivals are denoted by $S_1, S_2, S_3, \ldots$. They may be constant or of random duration. In the latter case, $\{S_1, S_2, S_3, \ldots\}$ is usually characterized as a sequence of independent and identically distributed random variables.
- In addition, in some systems, service times depend upon the time of day or upon the length of the waiting line.

Description of Queuing Sys (7/8)

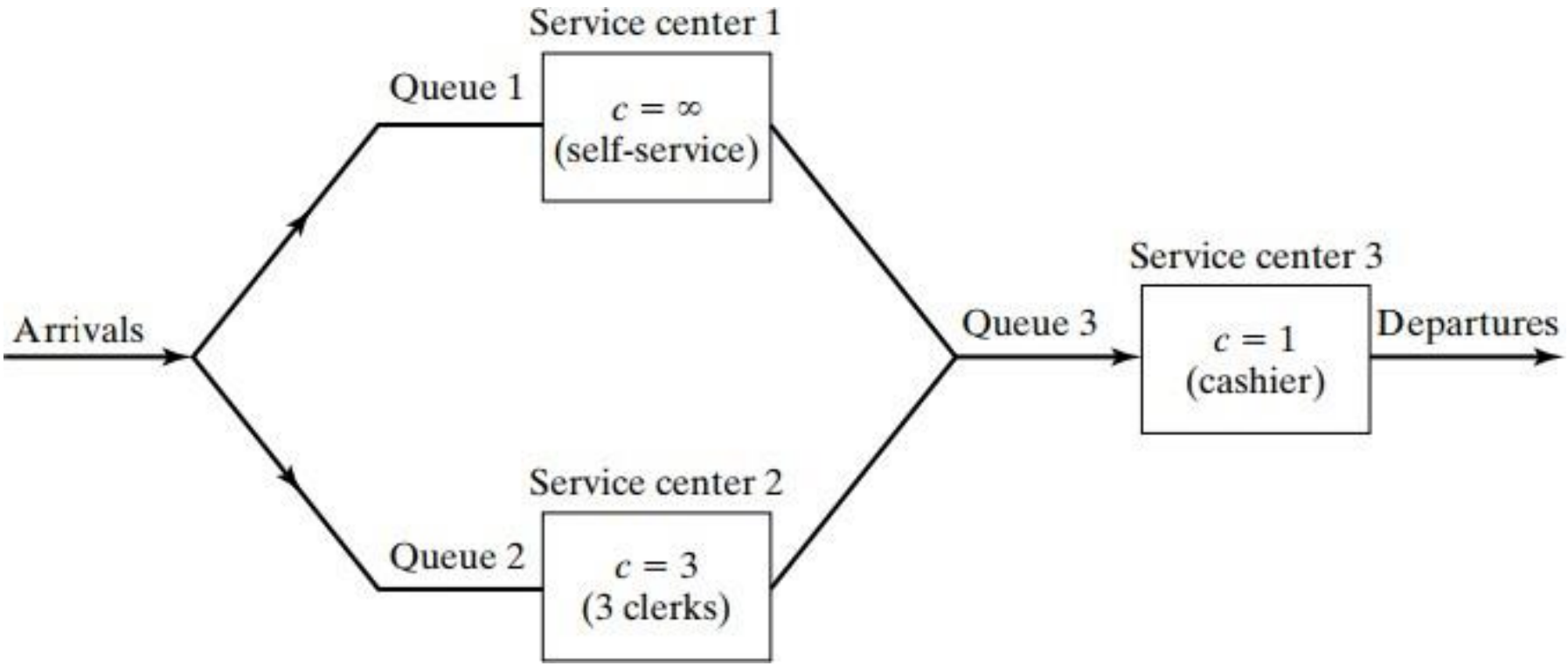
5. Service Times and the Service Mechanism (2/3):

- A queueing system consists of a number of service centers and interconnecting queues. Each service center consists of some number of servers, c, working in parallel; that is, upon getting to the head of the line, a customer takes the first available server.



Description of Queuing Sys (7/8)

5. Service Times and the Service Mechanism (3/3):



Description of Queuing Sys (7/8)

A notational system for parallel server systems: ***A/B/c/N/K***. These letters represent the following system characteristics:

A represents the interarrival-time distribution.

B represents the service-time distribution.

c represents the number of parallel servers.

N represents the system capacity.

K represents the size of the calling population.

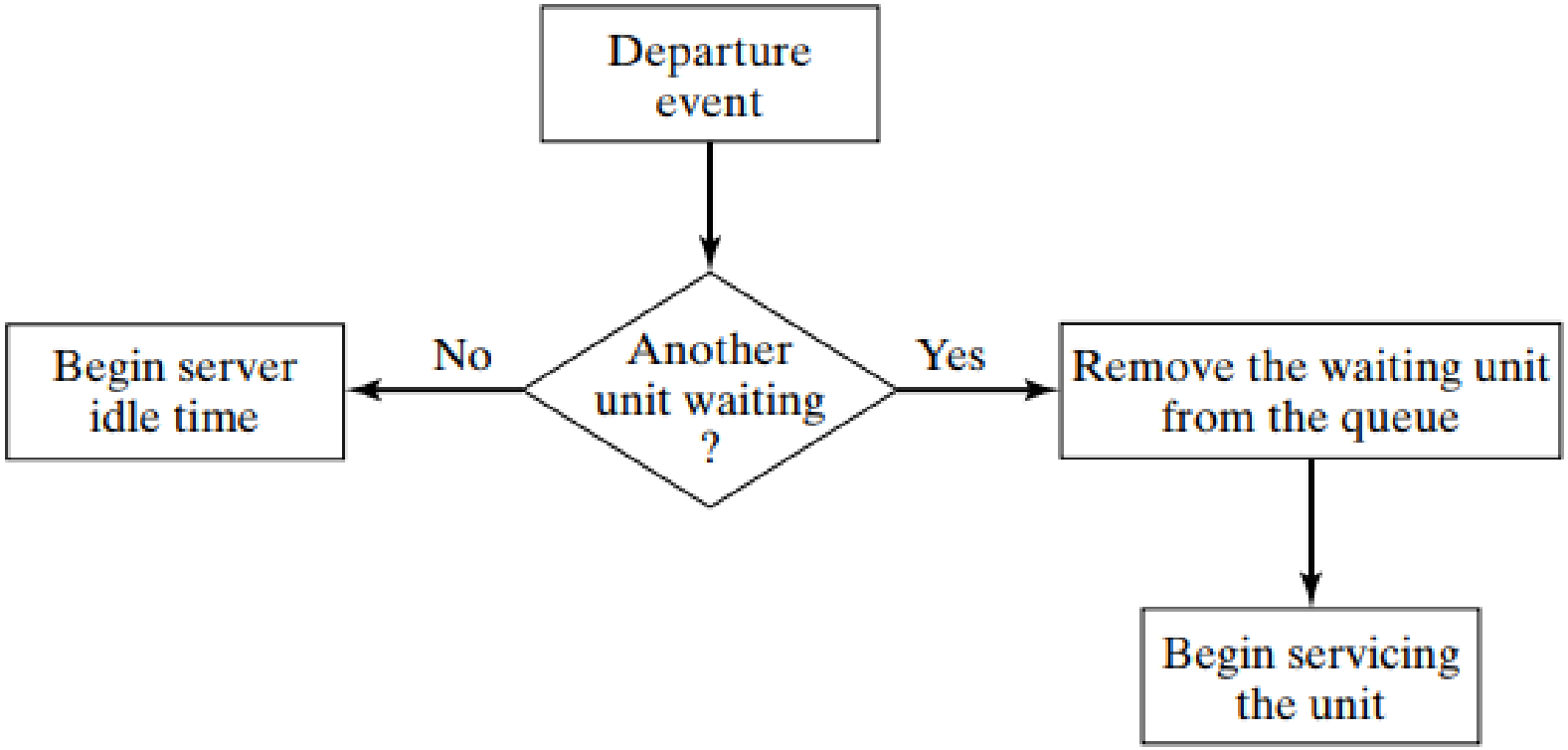
Description of Queuing Sys (8/8)

Queuing System State (1/3):

- The quantities (variables) that collectively completely describe the system at any given instant from the viewpoint of the study objectives:
  - **Number of units in the system:** How many units are currently in the queue or being processed by the server.
  - **Status of server (idle, busy):** Is the server(s) busy with processing some units or idle waiting for a job.

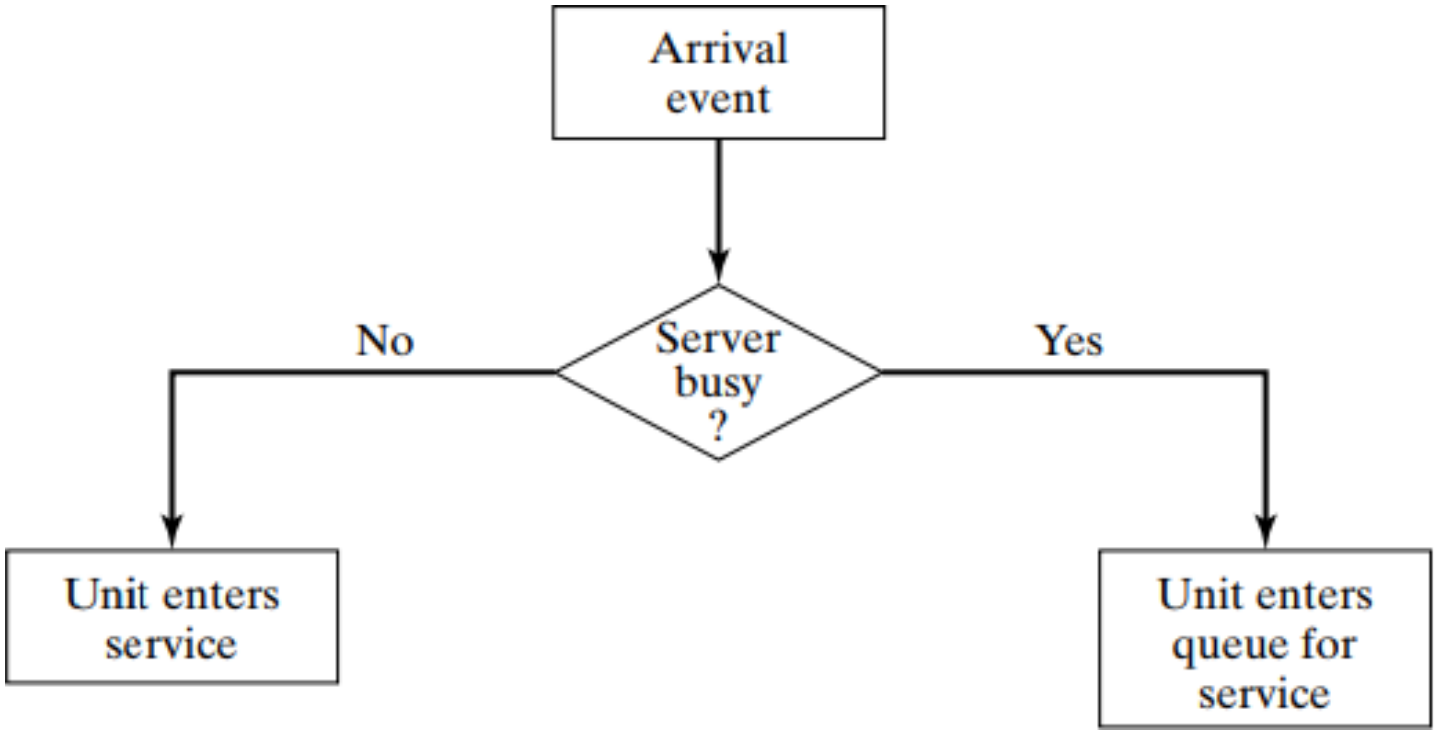
Description of Queuing Sys (8/8)

Queuing System State (2/3):



Description of Queuing Sys (8/8)

Queuing System State (3/3):



Thank You

Next Up:

- Single Server Queues
- Multiple Server Queues